

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

**PERFORMANCE ANALYSIS OF IMAGE ON
CRYPTOGRAPHIC ALGORITHMS**

BY
MA THAZIN KHAING
VI IT - 7

GRADUATION THESIS

OCTOBER, 2025
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**A THESIS
SUBMITTED TO THE DEPARTMENT OF
INFORMATION TECHNOLOGY ENGINEERING
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FOR THE DEGREE OF BACHELOR OF ENGINEERING
(INFORMATION TECHNOLOGY ENGINEERING)**

OCTOBER, 2025

THANLYIN

TECHNOLOGICAL UNIVERSITY (THANLYIN)
DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

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ABSTRACT

In this thesis, a system for Performance Analysis of Image on Cryptographic Algorithms is developed to evaluate the efficiency and security of symmetric encryption techniques in image data protection. The study focuses on four widely used algorithms—AES, DES, Blowfish, and Twofish—implemented to encrypt and decrypt image files while measuring key performance metrics such as elapsed time, throughput, memory consumption, and efficiency. The system is designed with a graphical interface that allows users to select an algorithm, process images, and visualize both encrypted outputs and performance results in real time.

The objective of this research is to identify the most suitable algorithm for image encryption tasks by comparing their computational performance and practical applicability. Results indicate that each algorithm demonstrates distinct strengths: AES provides the fastest processing, Blowfish offers the best balance of throughput and efficiency, DES performs moderately but is limited by security weaknesses, and Twofish shows poor suitability due to high processing overhead.

Furthermore, an Excel-based analysis module was integrated to enable structured data collection, ranking algorithms across different parameters, and generating visual reports for detailed evaluation. Experimental findings confirm that Blowfish is the strongest overall performer for bulk encryption, AES is optimal for speed-critical tasks, DES is primarily useful in legacy contexts, and Twofish is impractical for modern image encryption applications. This work contributes to both academic research and practical implementation by providing a systematic framework to evaluate cryptographic algorithms in multimedia applications. The study highlights the importance of choosing appropriate encryption methods based on performance trade-offs, ensuring data security while maintaining efficiency in real-world systems.

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